V-M-81c
OCTOBER 24, 1956
SUPERSEDING
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September 13, 1956 and
Fed. Spec. V–M–81b
December 15, 1948

FEDERAL SPECIFICATION

MATTRESSES, BED, FELTED, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers felted cotton mattresses for use on beds, cots, and bunks.

1.1.1 *Federal specification coverage.*—This specification covers only the types, grades, and sizes of the commodity generally purchased by the Federal Government, and does not include all types, grades and sizes which are commercially available.

1.2 Classification.

1.2.1 *Types and grades.* — Felted cotton mattresses shall be of the following types and grades, as specified (see 6.2):

Type I.—Plain box edge.
Type II.—Rolled box edge.
Type III.—Plain edge, not boxed.

Grade A.—All cotton (see 6.1.1).
Grade B.—75 percent linters and 25 percent cotton.
Grade C.—60 percent linters and 40 percent cotton.

1.2.2 *Sizes.*—Felted cotton mattresses shall be of the sizes shown in table I, as specified (see 6.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and Standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

T–T–871—Twine, Cotton, Wrapping.
T–T–931—Twine, Cotton, Mattress.
V–T–276—Thread, Cotton.
TT–I–542 — Ink, Marking, Indelible, Black, for Fabrics.
CCC–B–811—Burlap, Jute.
CCC–C–419—Cloth, Cotton, Duck, Plied-Yarns (Army and Numbered).
CCC–O–721—Osnaburg, Cotton.
CCC–T–351—Ticking, Mattress and Pillow.
DDD–S–751 — Stitches; Seams, and Stitching.
DDD–T–86 — Tape, Cotton; Bleached, Dyed, or Gray, General Use.
LLL–B–631—Boxes; Fiber Corrugated (for Domestic Shipment).
LLL–B–636 — Boxes, Fiber, Solid (for Domestic Shipment).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

JAN–P–108 — Boxes; Fiberboard, Corrugated and Solid (Military Type).

JAN–C–517—Cloth, Label, Cotton, Permanent-Coated.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

U. S. Department of Agriculture:

Pamphlet — Development and Use of Standards for Grade, Color, and Character of American Cotton Linters.

Miscellaneous Publication No. 310—The Classification of Cotton.

(Application for copies should be addressed to the U. S. Department of Agriculture, Cotton Branch, Production and Marketing Administration, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Standard sample.—When specified (see 6.2) the finished mattresses shall conform in quality, workmanship, and appearance to the standard samples established by the procuring activity.

3.2 Materials.

3.2.1 *Linters.*—Linters shall be new (not previously used), shall conform to United States standard grade No. 3 average for grade B mattresses, and to United States standard grade No. 3–middle for grade C mattresses (see U. S. Department of Agriculture pamphlet, "Development and Use of Standards for Grade, Color, and Character of American Cotton Linters").

3.2.2 *Cotton.*—Cotton for the mattresses shall be new (not previously used), and shall be not lower in grade than any of the following United States standard grades for cotton: ~~No. 7 or Low Middling, No. 7~~ or Low Middling Spotted, ~~No. 4 or Strict Middling Gray~~ (see U. S. Department of Agriculture Miscellaneous Publication No. 310). The use of reginned cotton will be permitted provided the reginned cotton is equivalent in grade to United States standard grades for cotton: ~~No. 7 or Low Middling, No. 7 or~~ Low Middling Spotted, ~~or~~ No. ~~4 or Strict~~ Middling ~~Gray~~. Tinged or stained cotton shall not be used.

3.2.2.1 *Grades A and C.*—Cotton for grades A and C mattresses shall have a staple length of not less than ½ inch.

3.2.2.2 *Grade B.* — Cotton for grade B mattresses shall have a staple length of not less than ⅞ inch.

3.2.3 *Felt.*—The felt shall be thoroughly dusted and well garnetted so as to produce unbroken laminated sheets of interlaced garnetted felt.

3.2.3.1 *Grade A.*—Felt for grade A mattresses shall be 100 percent cotton conforming to 3.2.2 and 3.2.2.1.

3.2.3.2 *Grade B.*—Felt for grade B mattresses shall be made of a blend of not more than 75 percent grade No. 3 average linters or better (see 3.2.1) and not less than 25 percent cotton conforming to 3.2.2 and 3.2.2.2.

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3.2.3.3 *Grade C.*—Felt for grade C mattresses shall be made from a blend of not more than 60 percent grade No. 3 middle linters or better (see 3.2.1) and not less than 40 percent cotton conforming to 3.2.2 and 3.2.2.1.

3.2.4 *Cloth.*—The cloth for mattress cases shall be of the following materials as specified, (see 6.2):

(a) Ticking conforming to type II of Federal Specification CCC–T–351, except that widths shall be as specified in 3.3 or as specified in the contract or order.

(b) Cotton duck conforming to type III of Federal Specification CCC–C–419 having a minimum weight of 9.85 ounces per square yard.

3.2.5 *Thread, sewing.*—The thread shall be unbleached cotton thread type IA1, 3 cord, ticket No. 30 conforming to Federal Specification V–T–276.

3.2.6 *Twine (for hand stitching of side, end, and roll-edge).*—The twine shall conform to type II of Federal Specification T–T–931.

3.2.7 *Tufting twine, braid, or tape.*—Cotton or flax tufting twine or tubular braid having a breaking strength of not less than 32 pounds or a single ⅝-inch cotton unbleached tape conforming to Federal Specification DDD–T–86 shall be used.

3.3 Mattress case.—The top and bottom sections of the mattress case material shall each be made up of not more than two longitudinal pieces, properly matched and sewn together. Cross seams will not be permitted. The blue stripe of ticking conforming to type II of Federal Specification CCC–T–351 shall run lengthwise of the mattress case. The distance between the seams forming the box edge shall be 2 ± ½ inches less than the thickness of the mattress when measured as specified in 4.2.3.3. The box edge, if separate, shall be formed of not more than four separate pieces. When more than one piece is used in constructing the box edge, the pieces shall be of such length that all joining seams shall be at a corner of the mattress and the pattern shall be properly matched. On type I and type III mattresses, corners shall be of the square box type. Corners on type II mattresses shall be slightly rounded to permit the use of a roll edge machine.

3.4 Stitches and seams.—The stitch and seam types specified hereinafter shall conform to Federal Specification DDD–S–751.

3.4.1 *Stitches.*—All machine stitching for seaming the mattress case material shall be made with stitch type 301 or 401 using 30/3 cord unbleached cotton thread with not less than 10 stitches per inch. Ends of all seams not caught in other seams or stitching shall be securely backstitched.

3.4.2 *Seams.*—The mattress case material shall be seamed with seam type LSb–2 or SSa–1. The closing end shall be seamed with seam type SSc–1.

3.5 Distribution of felt.—From 80 to 90 percent of the felt batts shall be distributed in layers of unbroken laminated sheets to produce uniform thickness. The remainder shall be uniformly and equally distributed over the middle one-third of the length of each face and the full width of the mattress to provide crowning.

3.6 Tufting. — Mattresses shall be well tufted on both sides with twine (see 3.2.7), tape, tying tufts, or tape with a corrosion-resistant metal locking strip, securely fastened to each end. The tying tufts shall be drawn through both sides of the mattress and knotted. Tufts shall be drawn on both sides to a length not more than the thickness of the mattress, when measured as specified in 4.2.3.3, minus 2 inches. In passing through the mattress case material the up-and-down

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twine or tape shall be sufficiently far apart to hold securely and avoid ripping and tearing of the intervening cloth. Tufts shall be given diamond spacing not more than 8 inches apart over the entire top and bottom surfaces.

Sizes X to 3, inclusive, shall have a minimum of 28 tufts. All other sizes shall have a minimum of 38 tufts.

3.7 Sizes.—Unless otherwise specified (see 6.2) dimensions and weights shall conform to table I for each size. Other sizes, when required, shall be of the length, width, thickness and weight specified (see 6.2) within the tolerances given in table I.

TABLE I.—*Sizes*

Size No.	Width ± ½ inch	Length ± ¼ inch	Thickness ± ¼ inch	Weight (Minimum)
	Inches	*Inches*	*Inches*	*Pounds*
X	26	72½	4	18
1	26	76	3	15
2	27	73	4	20
3	30	76	4	23
4	31	78	4	23
5	33	75	6	37
6	34	76	4	26
7	36	78	4½	31
9	38	75	6	42

3.8 Rolled edges and hand stitching.

3.8.1 *Rolled edges (type II only).*—The rolls shall be uniform in size and well filled with felt to form a compact, even roll. Stitching shall be 2 inches or less per stitch.

3.8.2 *Hand stitching (sides and ends).*—Two rows of single stitches through the boxing along each side and each end shall be equally spaced between the edges forming the boxing. The number of stitches shall be not less than those shown in table II. The number of stitches for sizes not tabulated shall be interpolated from table II.

TABLE II.—*Number of hand stitches*

Size No.	Stitches per single row, minimum	
	On each side	On each end
X	47	20
1	49	20
2	47	20
3	49	21
4	51	22
5	49	23
6	49	24
7	51	26
9	49	27

3.9 Labels.

3.9.1 *Nonmilitary agencies.*—Each mattress shall have a label attached to conform to the legal requirements of the state in which the mattress is manufactured. Each mattress also shall have a contractor's cloth label to comply with the requirements of each procuring activity.

3.9.2 *Military agencies.*—Each mattress shall have a label made of cloth conforming to Military Specification JAN–C–517. The marking shall be in indelible ink conforming to Federal Specification TT–I–542 and shall be in accordance with Military Standard MIL–STD–130, and shall include the size of mattress. In addition, each mattress shall have a label attached conforming to the legal requirements of the state in which the mattress is manufactured.

3.9.2.1 *For military medical procurement only.*—Mattresses shall have a label bearing the manufacturer's name or registered trademark and a caduceus approximately 1 inch in height centered between the letters "U" and "S" of appropriate size.

3.10 Workmanship.—Workmanship shall be first class in every respect. The use of oil in the fabrication of felt will not be permitted. The cotton felt shall be distributed in layers throughout the mattresses so as to produce a uniform thickness except for

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crowning and tufts, and side stitches shall be evenly spaced and drawn. The finished mattresses shall be free from defects affecting their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Lot.*—For purposes of sampling, a lot shall consist of not more than 3,000 mattresses of each type, grade, and size offered for delivery at one time.

4.1.2 *Sampling for visual and dimensional inspection.*—A random sample of mattresses shall be taken from each lot offered for Government inspection of visual and dimensional and weight characteristics in accordance with Military Standard MIL–STD–105 and inspection level III for lots of 25 and under, inspection level II for lots of 26 to 180, and inspection level I for lots of 181 and over. The A.Q.L. shall be equal to 2.5 percent defective.

4.1.2.1 *Sampling for internal inspection of mattresses.*—Where the inspector cannot observe the fabrication of the mattresses and conduct process inspection, a random sample of mattresses shall be taken from each inspection lot in accordance with table III, and mattresses shall be opened for internal inspection.

TABLE III.—*Sampling for internal inspection of mattresses*

Number of mattresses in lot	Number of mattresses in sample
300 or less	2
301 to 800	3
801 and over	5

4.1.3 *Sampling procedure for tests of materials.*—For the tests specified in 4.3, unless otherwise specified by the procuring activity, the manufacturer shall furnish, with each lot, test samples in the following quantities.

Ticking or duck ---------- 1 square yard.
Felt ---------- 1 pound.
Thread, sewing ---------- 50 yards.
Twine, hand stitching ---- 50 yards.
Tufting twine or braid ---- 15 yards.
Tufting tapes ---------- 10.

The manufacturer shall guarantee the test specimens furnished to be from the same material used in the construction of the mattresses in the lot.

4.2 Inspection.

4.2.1 *Visual, dimensional and weight inspection.*—Each of the sample mattresses taken in accordance with 4.1.2 shall be carefully inspected to determine conformance with this specification. Measurement of weight and dimensions shall be made as specified in 4.2.3. Any mattress in the sample containing one or more visual, dimensional or weight defects shall be rejected, and if the number of defective mattresses in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.2.2 *Inspection (internal).*—Each of the sample mattresses taken in accordance with table III shall be opened by the inspector and inspected internally for characteristics which cannot be inspected in the covered mattress. Nonconformance in any sample shall cause rejection of the lot.

4.2.3 *Methods of measurements and weight.*

4.2.3.1 *Felt.* — Unless otherwise specified (see 6.2), the cotton or linters, or both, shall be examined to determine conformance with 3.2.3 through 3.2.3.3 inclusive, by classifiers licensed by the U. S. Department of Agriculture; or representative samples each weighing not less than ½ pound, of the cotton or linters, or both, and of the felt may be forwarded to the administrative office concerned at Washington D. C., where inspection by the proper board of the U. S. Department of Agriculture will be obtained. The opinion of this board as to the true grade or quality will be final.

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4.2.3.1.1 *New cotton.*— In cases of dispute as to whether or not felt contains new cotton, a sample shall be examined under an ultra-violet light as follows:

A sample approximately 5 by 5 by 1 inch of the garnetted felt shall be prepared in such a manner as to expose a fresh surface. Comparison specimens of known history shall be similarly prepared. These shall include new unused cotton felt (less than 2 years old), old unused cotton felt (preferably more than 5 years old), and old used cotton felt (taken from used mattresses over 5 years old). The inspection sample and comparison specimens shall be exposed in a dark room for at least 5 minutes to ultra-violet radiation within the range of 3300 to 3900 angstrom units (see 6.4). The distance between the specimens and light filter shall be 3 to 5 inches. After irradiation, the fluorescence of the specimens shall be observed and compared. New unused cotton felt exhibits a uniform, purplish fluorescence over the entire surface with no background color evident. In old unused cotton, the surface fluorescence tends to diminish and a yellowish or ivory background predominates. Old used felt presents a somewhat stronger yellow background and may exhibit strong spotty fluorescence indicative of staining or contamination.

4.2.3.2 *Weight.* — Weight may be determined under prevailing atmospheric conditions, except in the settlement of disputes. The weight of the mattress shall then be determined after having been brought to equilibrium with an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of from 70° to 80°F. The material shall be considered to be in equilibrium when it shows no progressive change in weight.

4.2.3.3 *Measurement of thickness.*—Thickness shall be taken as the average of measurement at each of the four corners of a mattress made by determining the free height under the corner of an 18-inch square wood panel weighing 4 pounds, and resting on the mattress, with two edges over the adjacent edges of the mattress.

4.2.3.4 *Measurement of length and width.* Length and width shall be taken as the average of three measurements in each direction. Before measuring, mattresses shall be shaken out after removal from the bales and laid in a flat position.

4.3 Tests of material.

4.3.1 *Place of test.*—The specimens of material furnished with the lot as specified in 4.1.3 shall be tested at the place of manufacture if possible; otherwise, they shall be tested at a Government laboratory designated by the procuring activity. In the latter case, acceptance of the lot shall be withheld until a satisfactory test report on the samples representing that lot has been received by the inspector.

4.3.2 *Methods of test.*—The specimens shall be tested as specified in the referenced specifications to determine conformance with this specification.

4.3.2.1 *Tufting twine.*—The strength of the tufting twine shall be determined on a motor-driven machine equipped with grips of the drum type, not less than 1 inch in diameter. The grips through which the load is applied shall travel at a uniform rate of 12 inches ± ½ inch per minute. The distance between the centers of the drums at the beginning of the test shall be not less than 10 inches ± ¼ inch. Ten specimens from each sample shall be tested and the breaking strength shall be the average obtained from the 10 specimens. If, for any reason attributable to faulty operation, the strength of the specimens falls markedly below the average for the set, the results shall be disregarded and another specimen shall be broken.

4.3.3 *Rejection.*—If any one of the specimens fails under test, the entire lot of mattresses shall be rejected. A rejected lot may be resubmitted for Government inspection, but only after the manufacturer, subsequent to the original inspection, has completely re-

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worked the lot and repaired all nonconforming mattresses.

5. PREPARATION FOR DELIVERY

(For definitions and levels see 6.6.)

5.1 Packaging.

5.1.1 *Level A.* — Each mattress shall be individually packaged in a bag fabricated of polyethylene film of 0.004-inch thickness with no wall thickness falling below 0.00325-inch. All seams and closure shall be effected by a continuous heat seal. Excess air shall be removed from bags before complete closure is effected.

5.1.2 *Level C.*—Mattresses shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Unless otherwise specified in the contract or order, three mattresses of one type, grade, and size only, packaged by level A or C as specified (see 6.2), shall be packed flat and wrapped in a cloth covering material of cotton osnaburg weighing not less than 6.8 ounces per square yard conforming to Federal Specification CCC–O–721; or jute burlap 7½ ounces per linear yard conforming to Federal Specification CCC–B–811. The cloth covering shall be sewn with 16-ply cotton twine conforming to Federal Specification T–T–871, with approximately one stitch to the inch and every third stitch knotted. Five inches of surplus covering shall be gathered on each of the four corners and securely sewn into ears for handles.

5.2.1.1 Alternatively, materials specified in 5.2.1 may be procured in tubing form, and closures effected as specified in 5.2.1.

5.2.1.2 When specified by the procuring agency, three mattresses, packaged by level A or C as specified (see 6.2) shall be packed flat in a snug-fitting V-board fiberboard box, constructed, sealed, and strapped in conformance with Military Specification JAN–P–108, style FTC. Box dimensional and weight requirements of Military Specification JAN–P–108 shall be waived.

5.2.2 *Level B.*—Unless otherwise specified, mattresses shall be packed as specified in 5.2.1.

5.2.2.1 When specified by the procuring agency, three mattresses, packaged by level A or C as specified (see 6.2) shall be packed flat in a snug-fitting fiberboard box, style FTC, constructed and sealed in conformance with Federal Specifications LLL–B–631 or LLL–B–636, respectively. The fiberboard shall comply with the special requirements of the applicable specification and inside dimension limitations shall be waived.

5.2.3 *Level C.*—Mattresses shall be packed in a manner acceptable to common carriers for safe transportation to point of destination specified in shipping instructions, at the lowest transportation rate for such supplies.

5.3 Marking.

5.3.1 *Military agencies.* — All containers shall be marked in conformance with Military Standard MIL–STD–129. In addition, fiberboard and rigid type shipping containers shall be marked on at least two side faces with the following legend: "STOW FLAT—DO NOT STOW ON END."

5.3.2 *Nonmilitary agencies.*—Each shipping container shall be plainly marked in block, on two adjacent sides, with the following information:

CONSIGNEE NAME OF ARTICLE
BILL OF LADING USE NO HOOKS

6. NOTES

6.1 Intended use.

6.1.1 Grade A mattresses are intended for use as alternates during periods of emergency.

6.1.2 The sizes used by the various agencies of the Federal Government and the respective bed sizes and types of beds for which they are intended are shown in table IV.

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TABLE IV.—*Size of beds used*

Size No. (see table I)	Bed size-spring width x length, head to foot (inside)	Type of bed	Using agencies
	Inches		
X	27 x 75	Enlisted men's quarters ashore	Navy
1	26 x 76	Folding cots	Army—Air Force Navy Medical Service
2	27 x 73	Ambulance train bunks and cabinet beds	Army Medical Service and Federal Prisons
3	30 x 76	Small hospital beds	Federal Prisons
4	31 x 78	Bedsteads and pack beds	Army: Veterans Hospitals
5	33 x 75	Dormitory beds	Dept. of Interior
6	34 x 76	Hospital beds	Military Medical Service
7	36 x 78	Standard hospital beds and steel bunks	Navy, Marine Corps, Veterans Hospitals, and Department of Interior
9	38 x 75	Domestic beds	Dept. of Interior

6.2 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, grade, and size required (see 1.2 and 3.7).
(c) When standard sample is required (see 3.1).
(d) Cloth for mattress cases (see 3.2.4).
(e) When cotton and linters or both shall be examined (see 4.2.3.1).
(f) Level of packaging and packing required (see 5.1 and 5.2).
(g) When packing of mattresses in fiberboard boxes is required (see 5.2.1.2 and 5.2.2.1).
(h) Test samples (see 4.1.3).

6.3 Attention is called to the fact that size "X" has been added and that sizes 8, 10, 11, and 12, formerly covered by Federal Specification V–M–81b, have been eliminated.

6.4 The Hanovia Analytic Model Quartz Lamp equipped with a No. 6527c filter will fulfill the ultra-violet radiation requirements specified in 4.2.3.1.1. This apparatus may be obtained from the Hanovia Chemical and Manufacturing Company, Newark, N. J.

6.5 Transportation description.—The transportation description applicable to this item is:

Carload minimum weights: In cars 40 feet 9 inches or less in length, inside measurements, 10,000; 12,000; 15,000; 18,000; 22,000; 30,000 pounds.

In cars exceeding 40 feet 9 inches in length, inside measurements, 16,000; 18,000; 21,000; 24,000; 30,000; 40,000 pounds.

Truckload minimum weight 5,600 pounds.

6.6 Federal Standard No. 102, Preservation, Packaging, and Packing Levels, should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—Q M
Navy—Y MC Md
Air Force.

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